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HISTORY OF SCIENCE

Cold Short

The impurity that made most of Europe's iron ore worthless turned out to be the fuel that made it usable.

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Some time around 1875, in an evening chemistry class at the Birkbeck Institution in London, a teacher named George Chaloner made a remark that his students would have taken as a figure of speech. The man who eliminates phosphorus by means of the Bessemer converter, he said, will make his fortune [1], [2].

In the room was a clerk from a Metropolitan police court named Sidney Gilchrist Thomas, who was twenty-five and had no laboratory, no employer in the steel trade, and no obvious business with the problem. He took the remark literally. Ten years later he was dead at thirty-four, and the process that carries his name was making steel out of ore that the industry had written off as rubbish.

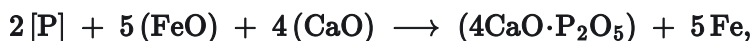
The scale of what had been written off is easy to understate. Henry Bessemer's converter, announced in 1856, could turn a charge of molten pig iron into steel in twenty minutes with no fuel but the impurities already dissolved in the metal — and it could do so only if that pig iron was nearly free of phosphorus. Most of the iron ore in Europe was not. The oolitic ironstones that ran in a belt through Lorraine, Luxembourg and the Cleveland hills of north Yorkshire carried enough phosphorus to make their pig iron useless for steel, and they were also the ores that lay in enormous, shallow, cheaply-worked beds. Britain's answer was to buy its ore abroad: imports rose from 208,000 tons in 1870 to 7,442,000 tons by 1913, the great bulk of it low-phosphorus haematite shipped from Bilbao [3]. That is a compound growth of nearly nine per cent a year, sustained for four decades, and all of it was the cost of one chemical constraint.

Forty Times Over

Phosphorus does not sit quietly in iron. It dissolves readily, then segregates to the grain boundaries as the metal solidifies, weakening exactly the surfaces along which a crack would like to run. Steel with too much of it fails brittly at ambient temperature — the trade's old word is *cold short*, the mirror image of the *hot shortness* that sulphur causes by forming low-melting sulphides at the same boundaries. Modern specifications cap phosphorus at between 0.03 and 0.05 per cent by mass, and above about 0.05 per cent it also drives centreline segregation in continuous casting. A Lorraine pig iron carried 1.8 to 2.2 per cent [4]. It was not marginal; it was forty times over.

The chemistry of the cure was not the mystery. Blow air through molten iron and the phosphorus oxidises along with everything else, to phosphorus pentoxide. The difficulty is what happens next. Phosphorus pentoxide is a strongly acidic oxide, and Bessemer's converter was lined with silica — itself an acid oxide. In an acid slag there is nothing for the pentoxide to bind to, so it sits at the interface and hands its oxygen back to the iron, returning the phosphorus to the bath. To hold it you must offer a base, and the obvious base is lime. But lime in a silica-lined vessel attacks the lining and destroys it.

That is the whole problem in one sentence: the cure and the container were chemically incompatible. Written properly, the reaction that removes phosphorus is



where square brackets mean dissolved in the metal and round brackets mean dissolved in the slag [5], [6]. It asks for two things at once — a strong base and plenty of iron oxide — and that double demand will come back at the end of this story.

George James Snelus had seen the answer first. Works chemist at Dowlais and later a Cumberland steelmaker, he found that lime burned hard enough became impervious to water, patented a lime converter lining in England in 1872, and then did not pursue it [7]. What Thomas added, working weekends and holidays with his cousin Percy Gilchrist — a chemist at the Blaenavon ironworks in Monmouthshire, and therefore the one person in the family with access to a furnace — was a lining that survived commercial use and a practice that made it work.

The lining was calcined dolomite. Dolomite is the double carbonate of calcium and magnesium, and calcining it drives off 47.7 per cent of its mass as carbon dioxide, leaving a mixed oxide that is 58 per cent lime and 42 per cent magnesia. The magnesia is what earns its place: it melts above 2,800 degrees Celsius and does not slake, so the lining resists both the heat and the water that would crumble pure burnt lime.

The Lime That Phosphorus Demands

The arithmetic of the basic process is unusually satisfying, because three numbers that were reported separately by people who had no reason to compare them all turn out to fit.

Start from the metal. A tonne of minette pig iron at 1.9 per cent phosphorus carries 19 kilograms of it, which oxidises to 43.5 kilograms of pentoxide, since the pentoxide weighs 2.291 times the phosphorus in it. Slagged as tetracalcium phosphate, each kilogram of phosphorus demands 3.62 kilograms of lime — so the phosphorus alone calls for 69 kilograms, before any is spent neutralising the silica.

Now come at it from the slag. The by-product of the Thomas process was ground and sold as fertiliser, which means it was assayed to death: it ran 14 to 18 per cent phosphorus pentoxide in citric-soluble form, on 45 to 50 per cent lime [8], [9]. Those two assays and the 43.5 kilograms fix the slag mass and the lime charge without any further information.

Slag grade, % P ₂ O ₅	Slag, kg per t	Lime at 47% CaO	Share of charge
14	311	146 kg	14.6 %
16	272	128 kg	12.8 %
18	242	114 kg	11.4 %

The reported operating practice was to charge lime at 12 to 15 per cent of the weight of the pig iron [4], [10]. The middle of the assay band lands squarely in the middle of the practice band, which is the sort of agreement that cannot be arranged after the fact. A second shop-floor rule decodes the same way: at least three kilograms of lime per 0.1 per cent of silicon per tonne of iron [11]. One per cent silicon yields 21.4 kilograms of silica, so thirty kilograms of lime against it is a lime-to-silica ratio of 1.40 — a working man's rule of thumb that is really a slag basicity in disguise.

THE LIME IS IN THE VESSEL FROM THE START OF THE BLOW. WHY DOES THE phosphorus only leave near the end?

Because the reaction needs iron oxide as well as lime, and for most of the blow there is almost none. Carbon dissolved in the bath is a far more eager reducing agent than iron, so as long as any of it remains, oxygen blown into the metal goes to making carbon monoxide rather than iron oxide. Only when the carbon is nearly gone — the moment the flame at the mouth of the converter drops, which is how the blowers knew — does iron oxide begin to accumulate in the slag, and only then does the phosphorus begin to move. The basic process therefore has a stage the acid process does not: the *afterblow*, a few extra minutes of blowing into metal that is already decarburised, spent entirely on the phosphorus [11].

Getting that timing wrong is presumably why Gilchrist's hundred-odd analyses at Blaenavon, on charges ranging from six pounds to ten hundredweight, recorded phosphorus removals anywhere from 20 to 99.9 per cent [1]. Done properly the figure is around 97 per cent: from 1.9 per cent in the pig to 0.05 in the steel. The slag that results carries 6.98 per cent phosphorus by mass against the metal's 0.05, a partition ratio of about 140 — a factor that has to be earned by basicity and iron oxide together, which is precisely what the reaction above says.

Which Impurity Is the Fuel

Here the story turns, and it turns on the fact that a Bessemer converter is not heated. It has no fuel line and no regenerator; its entire thermal budget is the heat of oxidising the impurities in its own charge. That makes the choice of which impurities to allow a thermodynamic decision as much as a chemical one.

Standard enthalpies of formation give the heat released per kilogram of each element burned. Silicon to silica yields 32.4 megajoules per kilogram, phosphorus to pentoxide 24.1, carbon to monoxide 9.2, manganese to monoxide 7.0. Carbon is taken only as far as monoxide deliberately: the second oxidation to dioxide happens in the flame above the bath, and that heat goes up the stack.

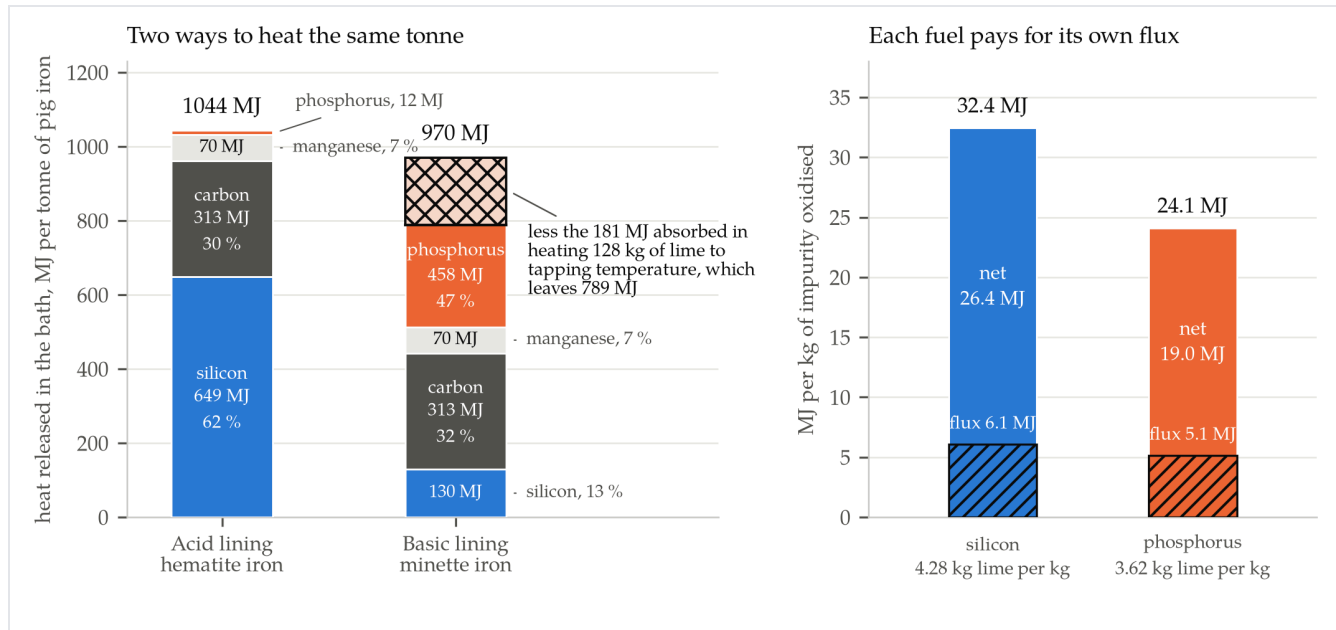


FIGURE Left: where the heat in a converter comes from, per tonne of pig iron, for a low-phosphorus haematite iron in an acid vessel and a Lorraine minette iron in a basic one. The totals are within seven per cent of each other, but the composition inverts — silicon supplies 62 per cent of the acid blow, phosphorus 47 per cent of the basic one, more than the carbon does. The hatched deduction is the heat absorbed in raising the 128 kg of lime the phosphorus demands to tapping temperature. Right: the flux bill against the heat released, per kilogram of impurity. Both remain net fuels after paying it, which is why the constraint that kept silicon out of the basic converter was refractory wear rather than arithmetic.

A haematite iron for the acid process runs about 2.0 per cent silicon, and silicon accordingly supplies 649 of its 1,044 megajoules a tonne — 62 per cent of the blow. Bessemer's process was, thermally speaking, a silicon fire.

Basic-process iron cannot be like that. Silica is what a dolomite lining is protected against, and every kilogram of silicon also demands 4.28 kilograms of lime to neutralise its oxide at a basicity of two, on top of the lime the phosphorus already wants. So the specification for Thomas iron pushes silicon down to a few tenths of one per cent — and with silicon gone, something has to take its place. Phosphorus does. At 1.9 per cent it delivers 458 megajoules a tonne: 47 per cent of the blow, and half again as much as the carbon.

The symmetry is exact and slightly absurd. Each version of the converter is forbidden the other's principal fuel — phosphorus by the chemistry of an acid lining, silicon by the wear it inflicts on a basic one — and each ends up running on the impurity the other cannot tolerate.

Nor is phosphorus a grudging fuel that barely covers its own overheads. The 128 kilograms of lime absorb about 181 megajoules on the way to tapping temperature, which is 19 per cent of the blow and leaves 789. Per kilogram of phosphorus that works out at 6.74 kilograms of lime and 9.55 megajoules of sensible heat, against 24.08 megajoules released. Phosphorus pays for its own flux two and a half times over.

What One Remark Was Worth

Thomas announced the result at a meeting of the Iron and Steel Institute in March 1878 and it drew almost no attention [1], [12]. The paper he and Gilchrist wrote that September for the autumn meeting was not read until the following May. What changed matters was E. W. Richards, manager of Bolckow Vaughan's works at Middlesbrough, who put a full-scale converter at their disposal; the trials of 4 April 1879, on Cleveland iron in the middle of the district the phosphorus problem had most obviously blighted, settled the question commercially in an afternoon [1].

The response abroad was quicker than the response at home, and in one case it preceded the demonstration. Joseph Massenez, technical manager at Hoerde near Dortmund, bought the German rights in 1878 — before the Middlesbrough trials — and with Rheinische Stahlwerke had the process running in Germany in 1879 [13]. By the end of the century a quarter of the world's steel was made this way, second only to the acid Bessemer [4].

Almost none of it was made in the countries that invented it. The United States had the Lake Superior ores, which are low in phosphorus, and therefore no motive; Britain, having proved the process, largely went to the open-hearth furnace instead [10], [14]. What the basic converter did was unlock the minette, and with it the industrial base of Germany, France, Belgium and Luxembourg [15]. German steel output passed Britain's in 1893 and by 1914 was more than double it [16]. It is difficult to think of another laboratory result whose main economic consequence landed so squarely on a country other than its author's.

Thomas received the Iron and Steel Institute's Bessemer Gold Medal in 1883, jointly with Snelus — the first time the Institute had awarded two medals at once for the same process, an unusually graceful way of settling a priority dispute [7]. He had been overworking for years, his lungs were failing, and neither a long sea voyage nor a winter in Egypt helped. He died in Paris on 1 February 1885, aged thirty-four, and left his estate to his sister Lilian with instructions that it be spent on doing good discriminatingly; she outlived him by more than half a century and spent it largely on the conditions of working women [2], [17].

The waste stream turned out to be a business of its own. At 272 kilograms per tonne of steel and 16 per cent phosphorus pentoxide, Thomas slag was a phosphate ore in its own right — leaner than rock phosphate but already ground, already calcined, and lying in a heap next to a railway. Sold as basic slag in English and Thomasmehl in German, its European consumption passed four million tonnes in 1928, about 58 per cent of it in Germany [18]. Four million tonnes at 16 per cent is 640,000 tonnes of

pentoxide; world phosphate rock production that year was on the order of nine and a half million tonnes at roughly 30 per cent, or 2.85 million tonnes of pentoxide [19]. On those figures the steel industry was supplying something close to a fifth of the world's phosphate fertiliser as a by-product of not throwing its slag away. The same tonnage read backwards implies about fifteen million tonnes of Thomas steel behind it, which is the right order for continental Europe at the time — the two figures check each other.

The Impurity That Would Not Follow

The claim you will most often see made for the basic converter is that it removed phosphorus and sulphur. It did not really remove sulphur, and the reason is the most elegant thing in the whole subject.

Both removals want the same basic slag, because both impurities form acidic or sulphidic species that need a base to hold them. But they want opposite atmospheres. Phosphorus removal, as the reaction above insists, needs iron oxide in the slag — a high oxygen potential. Sulphur removal needs the opposite: for sulphur to displace oxygen from lime and sit in the slag as a sulphide, the oxygen activity has to be low [20], [21]. A converter blowing air or oxygen through molten iron is, by construction, the most oxidising vessel in the works. It is thermodynamically well suited to one job and thermodynamically hostile to the other.

	Phosphorus	Sulphur
Failure it causes	cold short	hot short
Slag basicity needed	high	high
Oxygen potential needed	high	low
Where it is removed today	in the converter	before it

That is not a historical curiosity. It is why an integrated steelworks in 2026 still runs a dedicated hot-metal desulphurisation station in the gap between the blast furnace and the converter, treating the iron while it is still saturated with carbon and its oxygen activity is at its lowest [21]. The two impurities have to be taken out in different vessels because no single vessel can be both things at once, and Thomas's converter was the first to make that trade-off visible.

The lineage is otherwise unbroken. The Linz-Donawitz process, commercialised at Linz in November 1952 and Donawitz in May 1953, is a basic-lined converter blown with pure oxygen instead of air, and it makes roughly three-quarters of the world's steel [22], [23]. Thomas's lining and Thomas's lime are still in there.

What has gone is the fertiliser. Modern hot metal reaches the converter at about 0.1 per cent phosphorus rather than 1.9, because the ore is better, so the slag comes out at around 1.9 per cent phosphorus pentoxide — eight times leaner than Thomas slag, and far too lean to grind and bag. An industry that once supplied a fifth of the world's phosphate disappeared not through competition but through the improvement of its own raw material.

And the ore that started all this has not entirely surrendered. The high-phosphorus oolitic ironstones — at Aswan, across western Siberia, in the sedimentary iron deposits that make up a large share of China’s twenty billion tonnes of reserves — are still largely unworked, because the phosphorus in them is intergrown with the iron at a scale that no conventional beneficiation can separate, and the modern constraint sits upstream of the converter entirely [24], [25]. Chaloner’s remark, suitably updated, is still waiting for someone.

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